TOPIC 7 | APPOINTMENT OVERBOOKING AND NO-SHOW MANAGEMENT

Overbooking Clinic Appointments with Reinforcement Learning

Exploration Schedules, Learning Efficiency and Robustness to Shifts in the No-Show Rate

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Problem Context: Patient No-Shows and Overbooking

- Patient no-shows leave scheduled clinic capacity unused; in the dataset studied here, 20.2% of appointments were missed.
- Clinics therefore overbook, issuing more appointments than there are regular slots in a session.
- Booking too few additional patients leaves slots idle and lengthens the waiting list, whereas booking too many causes overtime when most patients attend.
- The appropriate number of additional bookings depends on the patients scheduled, and therefore varies from one session to the next.
- Because each decision also affects the waiting list for later sessions, the task is a sequential decision problem and is addressed with reinforcement learning (RL) [3].

20.2%

of the 110,487 recorded appointments were missed

0 to 4

additional patients that the clinic may book before each session

Problem Statement and How It Is Addressed

Requirement	How this work addresses it
Dataset: Medical Appointment No Shows	All reported results are computed on the public Kaggle table of 110,487 appointments recorded in 2016 [17].
State: slots filled and predicted no-show risk band	Slots filled (3 bands) × risk band predicted by a calibrated classifier (4) × waiting list band (5), giving 60 states.
Action: overbooking level	Between 0 and 4 additional patients are booked before each session (5 actions).
Reward: utilisation less waiting and overtime	shows − 3 × overtime − 0.5 × idle slots − 0.4 × waiting list + 0.3 × patients admitted from the waiting list, as in Paper 7.
Futuristic angle: adaptation to patient behaviour	The policy responds to the predicted risk of each session and to the waiting list, and is evaluated under shifts in the no-show rate.
Evaluation: utilisation, waiting, overtime, convergence of MC and Q-learning	All four measures are reported on scoring weeks never used for learning (slides 12 to 17).

Paper 7: The Baseline Study

Scope of the original study

- Formulated overbooking as a Markov decision process (MDP) with 100 states and 5 actions.
- Evaluated all methods on a simulator specified by hand that reproduces the columns of the dataset, as the Kaggle site could not be accessed at the time.
- Learned policies from 150 logged clinic days using dynamic programming (DP), Monte Carlo (MC) control and Q-learning.
- Compared five exploration schedules and examined robustness to a change in the no-show rate.

Reported returns (simulator)

Method	Return
Guideline rule	160.3
DP on estimated MDP	162.2
Monte Carlo control	166.6
Q-learning	164.4
Oracle DP (10x data)	179.0

- **Identified gap:** the learned policies improved on the guideline rule by only +1.9 to +6.3, whereas DP with ten times as much data improved on it by +18.7; no real data were used.
- **Objective of this work:** to evaluate Paper 7's methods on the real dataset, to propose a method that outperforms them, and to report both in the same paper.

Related Work

Appointment scheduling and overbooking

- Appointment systems, no-shows, overbooking and the balance between waiting and overtime are well studied in operations research [1, 2].
- Predicted no-show probabilities have been combined with single session overbooking [6], and fixed risk thresholds for double booking are the baseline for learned policies [7].

Exploration in reinforcement learning

- Epsilon greedy exploration, decaying schedules and optimistic initial values are the standard devices [3]; R-max makes optimism explicit [5].
- Q-learning converges under conditions on its step size [8, 9]; other methods need schedules that are greedy in the limit with infinite exploration [4].

Offline RL, shift and structure

- Learning from logs is limited by data coverage and sensitive to model error, which motivates pessimism and support constraints [10, 11, 12].
- In a factored MDP the transition model is assembled from separately specified parts of the state [13]; this is the idea behind the proposed structured model.

- **Position of this work:** not a new scheduling model, but a study of how RL learners behave on real logged data, and of how much they gain from a model that knows the clinic's rules.

Background: Markov Decision Processes

Definition

- A finite MDP is a tuple (S, A, P, R, γ): states, actions, transition probabilities P(s′ | s, a), expected rewards R(s, a) and a discount factor γ in [0, 1).
- A policy π maps each state to an action; its value is the expected discounted return obtained from each state when π is followed [3, 14, 15].
- The optimal value V* satisfies the Bellman optimality equation, and an optimal policy acts greedily with respect to it.

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V^*(s') \right]$$

MDPs estimated from logged data

- Clinical data are logged: transitions (s, a, r, s′) are observed, and counts give $\hat{P}(s' \mid s, a) = N(s, a, s') / N(s, a)$.
- Solving the estimated MDP by DP gives a policy directly; MC and Q-learning learn from experience sampled from it, without involving patients.
- Pairs observed fewer than 10 times are treated pessimistically, because planning on an estimated model is easily misled by its errors [10, 11, 12].

Background: The Three Solution Methods

Dynamic programming (DP)

- Requires the model (P, R) and computes V^* as the fixed point of the Bellman equation.
- Value iteration repeats the Bellman update until the change falls below a tolerance.
- Policy iteration alternates exact evaluation with greedy improvement; both give policies of equal value here.

Monte Carlo (MC) control

- Model free: estimates Q(s, a) as the average return after first visits to (s, a) in complete episodes [3, 16].
- Improves a policy that is greedy with respect to Q while still exploring with probability ε.
- Unbiased but of high variance, and learns only once an episode has ended.

Q-learning

- Model free temporal difference method that bootstraps from its own estimates [8].
- Converges to Q^* if every pair is visited infinitely often and the step sizes decay suitably.
- Spreads reward information faster than MC, at the cost of bias from bootstrapping on noisy estimates.

$$\textit{Q-learning update:}\quad Q(s, a) \leftarrow Q(s, a) + \alpha\,[\, r + \gamma \max_{a'} Q(s', a') - Q(s, a)\,]$$

In this work all three methods learn from the same 150 logged clinic days; MC and Q-learning sample experience from the estimated MDP, never from the scoring sessions.

The Medical Appointment No Shows Dataset

Dataset characteristics

- 110,487 appointments of 62,277 patients across 81 neighbourhoods in Brazil, recorded on 26 weekdays between 29 April and 8 June 2016 [17].
- The overall no-show rate is 20.2%, and it varies little by weekday (19.4% to 21.2%).
- Booking lead time is the strongest predictor: 4.7% of same day bookings were missed, compared with 33.0% of bookings made 31 or more days in advance.
- Before the analysis, one record with a negative age and the 39 appointments held on Saturdays were removed.

No-show classifier

- Gradient boosting [18], calibrated by isotonic regression [19] and implemented in scikit-learn [20].
- Uses only information known at booking: lead time, age, sex, SMS reminder, health and welfare flags, weekday, history and neighbourhood.
- Trained on appointments before 9 May, calibrated on 9 to 15 May and applied from 16 May onwards, so no later information is used.
- Test AUC of 0.728; the mean predicted no-show rate is 22.0% against 19.8% observed.

Formulating the Problem as a Markov Decision Process

Session

All appointments of one neighbourhood on one day (at least five), with 10 regular slots.

Episode

40 sessions drawn from the real sessions while preserving the weekday order.

State

Slots filled (5 to 7, 8 to 9, 10) × predicted risk quartile (4) × waiting list band (5) = 60 states.

Action

The number of additional patients booked before the session: 0, 1, 2, 3 or 4.

Reward

shows − 3 × overtime − 0.5 × idle − 0.4 × waiting list + 0.3 × patients admitted from the waiting list.

Data split

Learning: 16 to 25 May (506 sessions). Scoring: 30 May to 8 June (592 sessions), unseen in learning.

Attendance of every booked patient is taken directly from the records; policies are learned from 150 logged clinic days, as in Paper 7.

Methods Compared

Methods from Paper 7

- DP (value iteration) on the MDP estimated from the logged data.
- Monte Carlo control and Q-learning on the same estimated MDP.
- The guideline rule (one or two additional bookings) and the clinician behaviour policy.
- Oracle DP: DP fitted to 1,500 logged days, ten times the standard amount, as a reference.

Baselines from published work

- Single session optimiser: chooses the number of additional bookings that maximises the expected reward of the current session [6].
- Threshold double booking ($\theta = 0.25$): one additional booking for each patient whose predicted risk is at least θ [7].
- Fixed booking: always books $k = 2$ additional patients.

Evaluation protocol

- Each method is evaluated over 20 replicates (independent logged datasets) and 1,000 episodes of the scoring sessions with common random numbers; the return is the discounted reward over 40 sessions.

Proposed Method: A Structured Model of the Clinic

Motivation

- Paper 7's methods estimate every transition probability P(s' | s, a) from counts: 18,300 quantities from 150 days.
- Booking three or four additional patients is rare in the logs, so the estimates for these actions are unreliable.
- Most of the clinic's dynamics are known in advance: the slots, the reward and how additional bookings reduce the waiting list.

Procedure

1. Estimate only patient behaviour from the logs: cell frequencies (12), the waiting list arrival rate (1) and waiting list lengths within each band (13).
2. Compute the expected reward of each session exactly from the patients' predicted attendance probabilities.
3. Assemble the transition model from these components, a factored MDP [13].
4. Solve it by value iteration (DP); MC and Q-learning can also be trained on it.

26

parameters in the structured model

18,300

transition probabilities in the estimated MDP

- Every logged session informs all of the structured model's parameters, which is why it needs far less data than the estimated MDP.

Main Results on the Real Scoring Sessions

Method	Return (± 95% CI)	Utilisation	Overtime	Waiting list
Paper 7: Guideline rule	150.8	88.9%	0.20	5.28
Threshold double booking (θ = 0.25)	164.5	95.5%	1.02	1.76
Fixed booking (k = 2)	174.6	91.4%	0.37	2.45
Single session optimiser	182.0	93.8%	0.51	1.83
Paper 7: DP on estimated MDP	170.3 ± 0.99	90.4%	0.39	2.46
Paper 7: Monte Carlo control	162.8 ± 2.26	89.2%	0.38	2.86
Paper 7: Q-learning	167.0 ± 2.11	90.1%	0.42	2.53
Paper 7: Oracle DP (10x data)	173.6	91.5%	0.54	1.72
Monte Carlo, structured model	169.3 ± 1.70	92.4%	0.65	2.00
Q-learning, structured model	175.2 ± 1.05	92.8%	0.59	1.81
Structured model DP (ours)	**182.5 ± 0.01**	**93.8%**	**0.53**	**1.67**

182.5

return of the structured model, the highest of all methods

+12.3

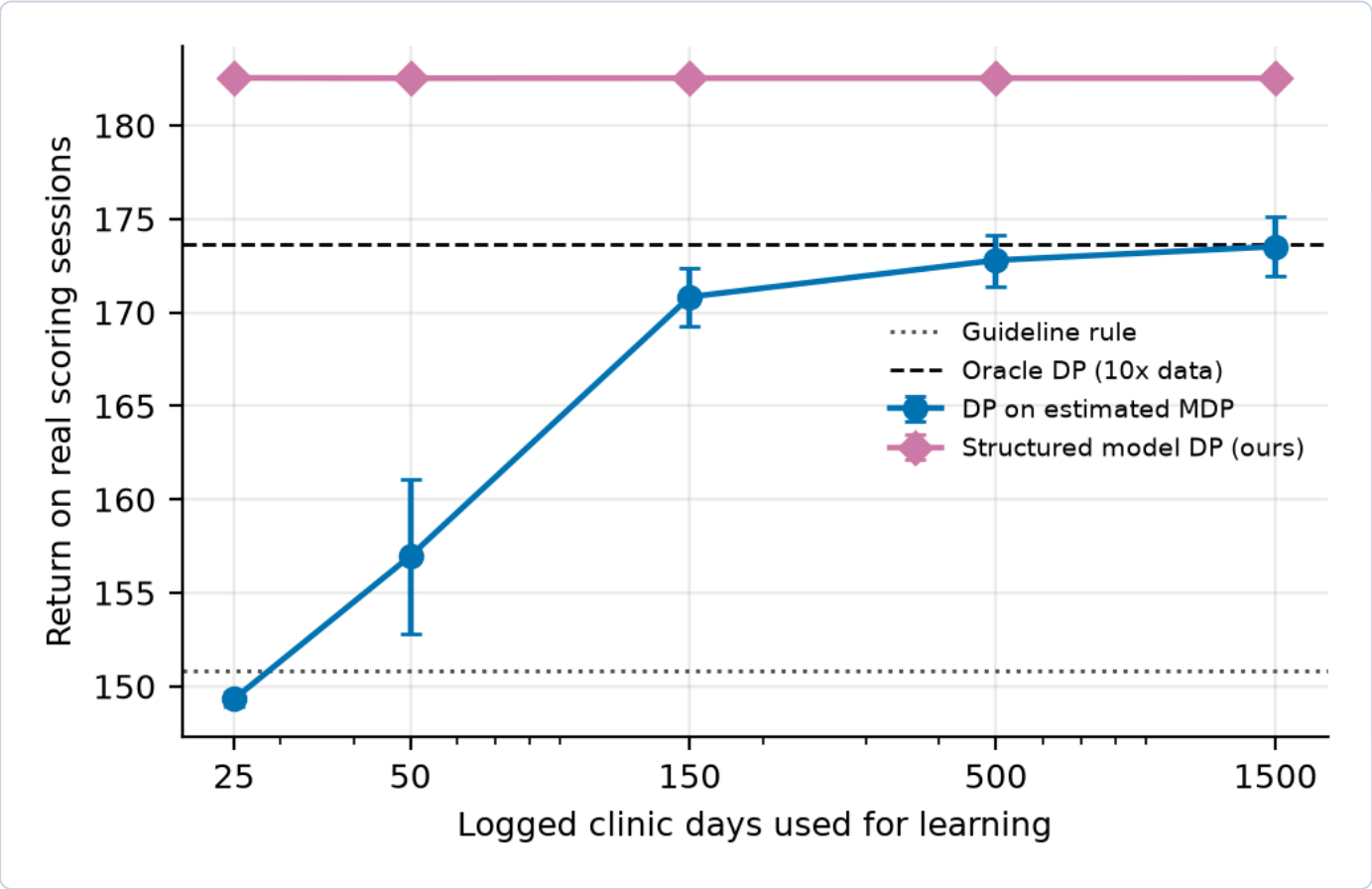
over DP, the best of Paper 7's methods ($p < 0.001$)

+0.51

over the strongest published baseline ($p < 0.001$)

Return: discounted reward over 40 sessions (mean ± 95% CI over 20 replicates). Overtime and waiting list are per session.

Effect of the Amount of Logged Data



- With 25 logged days, DP on the estimated MDP attains 149.3, below the guideline rule (150.8); with 1,500 days it reaches 173.5.
- The structured model attains 182.5 at every data size from 25 to 1,500 days.
- The reason is the number of quantities estimated: 26 parameters instead of 18,300 transition probabilities.
- The limitation of Paper 7's methods is therefore the coverage of the logged data, not the learning algorithm.

Robustness to Changes in the No-Show Rate

Method	Nominal rate (18.5%)	Higher rate (26.4%)	Lower rate (13.5%)
Guideline rule	150.8	129.3	157.4
DP on estimated MDP	170.3	155.6	171.5
Monte Carlo control	162.8	147.7	165.6
Q-learning	167.0	153.4	168.1
Oracle DP (10x data)	173.6	164.1	171.5
Single session optimiser	182.0	172.0	176.8
Structured model DP (ours)	**182.5**	**173.2**	**177.4**

Observations

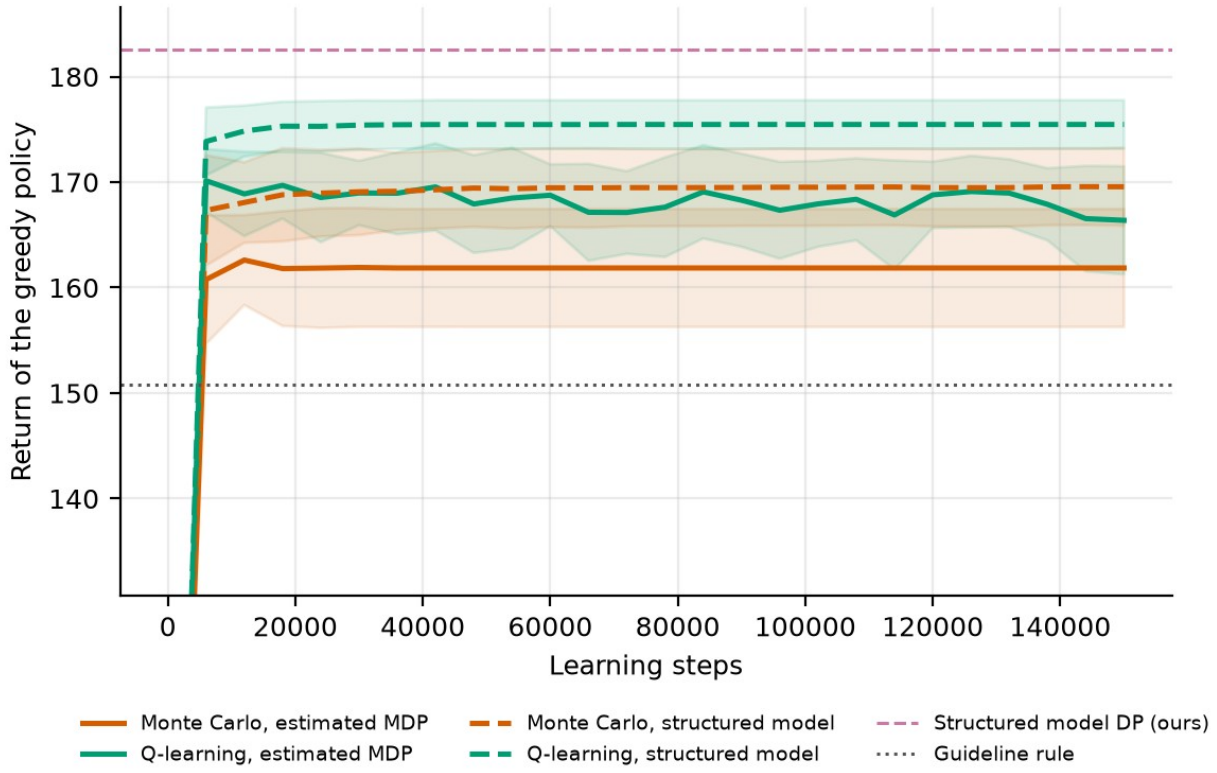
- The structured model has the highest return at all three rates.
- Rising no-shows affect the guideline rule most: its return falls from 150.8 to 129.3.
- Policies are not retrained; only the recorded attendance outcomes are altered.

Benefit of Planning Across Sessions Under Higher Demand

Waiting cost	New patients per session	DP on estimated MDP	Best fixed booking	Single session optimiser	Structured model	Gain
0.4	1.5	170.6	174.5	181.6	182.0	+0.2%
1.0	1.5	144.0	149.2	155.3	157.9	+1.7%
0.4	2.5	111.8	160.1	160.0	168.4	+5.3%
1.0	2.5	42.3	110.7	94.6	126.1	+33.3%

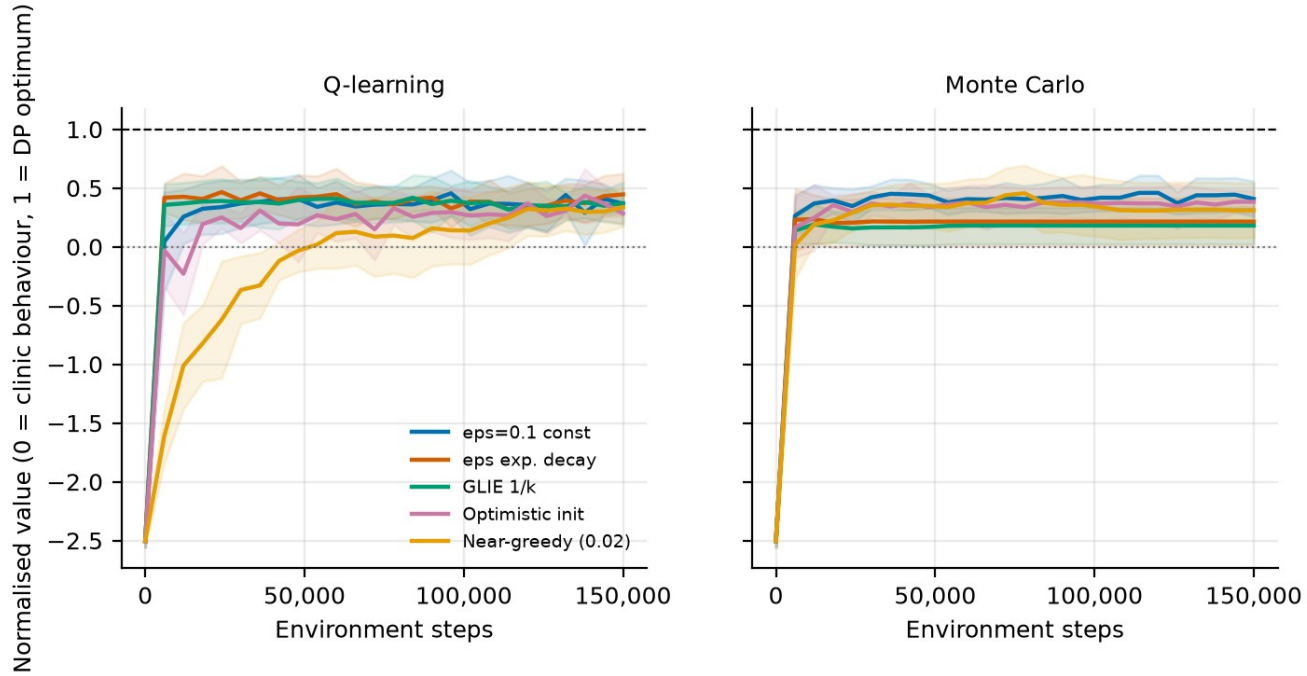
- The single session optimiser considers only the current session, whereas the structured model also accounts for the effect of each decision on the waiting list in later sessions.
- This advantage grows with demand: the gain over the single session optimiser rises from +0.2% to +33.3%.
- DP on the estimated MDP deteriorates sharply in the busiest setting (42.3), because the actions it requires are rarely observed in the logs.
- A waiting cost of 0.4 with 1.5 new patients per session corresponds to Paper 7's setting (slide 12).

Convergence of Monte Carlo Control and Q-learning



- On the estimated MDP, MC converges to a return of 161.9 and Q-learning to 166.4.
- On the structured model, Q-learning reaches 175.5 and MC 169.6, against 182.5 for its DP solution.
- Q-learning approaches the DP solution more closely than MC on both models.
- On the structured model, both learners settle within 18,000 of the 150,000 learning steps.

Exploration Schedules



- Five schedules were compared for MC and Q-learning: constant ε = 0.1, exponential decay, a 1/k decay, near greedy (ε = 0.02) and optimistic initialisation.
- Learning curves are normalised so that 0 is the clinic's behaviour and 1 the optimum, and summarised by their mean value (AUC).
- No run reached 90% of the possible improvement (0 of 60): the learners level off near the fitted model's DP solution (126.6), below the optimum (144.9).
- For Q-learning, near greedy exploration was the slowest (AUC -0.16 against 0.29 for exponential decay).
- Exploration inside a fitted model cannot reveal actions absent from the logs, so the data, not the schedule, limit performance.

Implementation and Reproducibility

Software pipeline (Python)

- **noshow_data.py:** data cleaning, classifier training and construction of sessions.
- **real_clinic.py:** the clinic MDP built from the real sessions.
- **planner.py:** the structured model proposed in this work.
- **rl_core.py, common.py, ideas.py:** Paper 7's solvers and experiments, reused without modification.
- **run_real.py:** executes all experiments and writes results.json and the figures.
- **paper07_v3.py, paper07_v4.py:** generate the paper from results.json; check scripts verify it.

Models, tools and reproducibility

- Gradient boosting classifier with isotonic calibration (scikit-learn).
- Value iteration, Monte Carlo control and Q-learning from Paper 7's rl_core module.
- numpy, scipy, matplotlib and python-docx.
- **Code:** github.com/Nagasai-Datta/ml_lab, folder topic7.
- **Reproduction:** python topic7/code/run_real.py reruns every experiment with fixed random seeds.

Limitations

- Only attendance is observed in the data; the overbooking decisions in the logs are generated and the waiting list is simulated.
- The structured model assumes that the clinic's rules (slots, reward weights and waiting list dynamics) are known.
- Only about 9% of real sessions have fewer than 10 bookings, so the slots filled component of the state rarely varies.
- The no-show rate drifted from 21.2% to 18.5% between periods; the classifier predicts 22.0% against 19.8% observed.
- Waiting is measured by the length of the waiting list, as the dataset records no times within a day.
- The published baselines are simplified implementations whose settings were tuned on the scoring sessions, which favours them.
- The findings concern algorithms rather than clinical practice, and no clinical claim is made.

Conclusions and Future Work

Conclusions

- On the real sessions, Paper 7's DP, MC and Q-learning outperform the guideline rule (170.3, 162.8 and 167.0 against 150.8) but remain below the oracle (173.6), which shows that the logged data are the limiting factor.
- Learning only patient behaviour removes this limitation: the structured model attains 182.5 and remains the best method when the no-show rate rises or falls, with as few as 25 logged days.
- Its advantage over the strongest published baseline increases from +0.2% to +33.3% as pressure on the waiting list grows.

Future work

- Collect logs in which the number of additional bookings is varied deliberately.
- Measure overtime and waiting from recorded visit times.
- Estimate the reward weights and waiting list dynamics from clinic records rather than assuming them.

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